~ SCREENSHOTZ

DSA-1 C++ CASE STUDY BY AKHILA

FIREGUARD: CLOUD SECURITY PACKET FILTERING SYSTEM

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard_Menu_Driven

  ╔══════════════════════════════════════════════════════════╗
  ║         FireGuard — Cloud Security Firewall v2.0         ║
  ║    Real-time Packet Inspection & Threat Prevention       ║
  ╚══════════════════════════════════════════════════════════╝
  [INIT] Default attack signatures loaded.
  [INIT] FireGuard is ACTIVE. All traffic will be monitored.

  ┌─────────────────────────────────────────────────┐
  │  MAIN MENU                                      │
  ├────┬────────────────────────────────────────────┤
  │ 1  │ Pattern List    — Manage attack signatures │
  │ 2  │ Logging Trail   — Audit log & retrace      │
  │ 3  │ Verification Line — Packet queue & inspect │
  │ 4  │ Socket Identity — Socket registry          │
  │ 5  │ Flow Sorter     — Data volume analysis     │
  │ 6  │ Firewall Interfaces — Topology map         │
  │ 7  │ Validation Path — Shortest bypass route    │
  │ 8  │ Traffic Allocator — Load balancing         │
  │ 0  │ Exit                                       │
  └────┴────────────────────────────────────────────┘
  Choice: 1

  [1] Pattern List
  a) Add pattern   b) Test payload   c) Show info
  > a
  Enter attack keyword/pattern: ransomware
  [PATTERN] "ransomware" added to signature database.
```

```
┌──────────────────────────────────────────────────────────┐
│  MAIN MENU                                               │
├─────┬────────────────────────────────────────────────────┤
│ 1   │ Pattern List    — Manage attack signatures         │
│ 2   │ Logging Trail   — Audit log & retrace              │
│ 3   │ Verification Line — Packet queue & inspect         │
│ 4   │ Socket Identity — Socket registry                  │
│ 5   │ Flow Sorter     — Data volume analysis             │
│ 6   │ Firewall Interfaces — Topology map                 │
│ 7   │ Validation Path — Shortest bypass route            │
│ 8   │ Traffic Allocator — Load balancing                 │
│ 0   │ Exit                                               │
└─────┴────────────────────────────────────────────────────┘
Choice: 1

[1] Pattern List
a) Add pattern   b) Test payload   c) Show info
> b
Enter payload to test: User attempted exec(command)
[ALERT] ⚠ THREAT DETECTED: matched pattern "exec("
```

```
┌────────────────────────────────────────────────────┐
  MAIN MENU
├─────┬──────────────────────────────────────────────┤
  1   │ Pattern List    — Manage attack signatures
  2   │ Logging Trail   — Audit log & retrace
  3   │ Verification Line — Packet queue & inspect
  4   │ Socket Identity — Socket registry
  5   │ Flow Sorter     — Data volume analysis
  6   │ Firewall Interfaces — Topology map
  7   │ Validation Path — Shortest bypass route
  8   │ Traffic Allocator — Load balancing
  0   │ Exit
└─────┴──────────────────────────────────────────────┘
Choice: 1

[1] Pattern List
a) Add pattern   b) Test payload   c) Show info
> c
Trie-based multi-pattern engine active.
Matching is O(m) per payload where m = payload length.
Similar to Palo Alto App-ID & Content-ID engines.
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard_Menu_Driven

Choice: 2

[2] Logging Trail
a) Retrace last N entries   b) Manual log entry
> b
Packet ID: PKT-001
Rule: DEEP_INSPECTION
Result (ALLOWED/BLOCKED/SUSPICIOUS): ALLOWED
[LOG] PKT-001 | Rule: DEEP_INSPECTION | Result: ALLOWED

┌──────────────────────────────────────────────────┐
│  MAIN MENU                                       │
├────┬─────────────────────────────────────────────┤
│ 1  │ Pattern List    — Manage attack signatures  │
│ 2  │ Logging Trail   — Audit log & retrace       │
│ 3  │ Verification Line — Packet queue & inspect  │
│ 4  │ Socket Identity — Socket registry           │
│ 5  │ Flow Sorter     — Data volume analysis      │
│ 6  │ Firewall Interfaces — Topology map          │
│ 7  │ Validation Path — Shortest bypass route     │
│ 8  │ Traffic Allocator — Load balancing          │
│ 0  │ Exit                                        │
└────┴─────────────────────────────────────────────┘
Choice: 2

[2] Logging Trail
a) Retrace last N entries   b) Manual log entry
> a
How many entries to retrace? 1

══ AUDIT TRAIL (last 1 entries, newest first) ══
Time      Packet ID       Rule Applied            Result
------------------------------------------------------------
22:14:03  PKT-001         DEEP_INSPECTION         ALLOWED
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard_Menu_Driver

┌──────────────────────────────────────────────────┐
│  MAIN MENU                                       │
├────┬─────────────────────────────────────────────┤
│ 1  │ Pattern List    — Manage attack signatures  │
│ 2  │ Logging Trail   — Audit log & retrace       │
│ 3  │ Verification Line — Packet queue & inspect  │
│ 4  │ Socket Identity — Socket registry           │
│ 5  │ Flow Sorter     — Data volume analysis      │
│ 6  │ Firewall Interfaces — Topology map          │
│ 7  │ Validation Path — Shortest bypass route     │
│ 8  │ Traffic Allocator — Load balancing          │
│ 0  │ Exit                                        │
└────┴─────────────────────────────────────────────┘
Choice: 3

[3] Verification Line (Packet Queue)
a) Enqueue packet   b) Inspect next   c) Inspect all   d) Queue size
> c
Inspecting all 2 packet(s)...

—— Inspecting Packet PKT-001 ——
[LOG] PKT-001 | Rule: DEEP_INSPECTION | Result: ALLOWED
[PASS] Packet PKT-001 is clean → ALLOWED

—— Inspecting Packet PKT-002 ——
[LOG] PKT-002 | Rule: PATTERN_MATCH:union select | Result: BLOCKED
[ALERT] Malicious payload detected: "union select" → BLOCKED
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard_Menu_Driven

  ┌──────────────────────────────────────────────────┐
  │ MAIN MENU                                        │
  ├────┬─────────────────────────────────────────────┤
  │ 1  │ Pattern List     – Manage attack signatures │
  │ 2  │ Logging Trail    – Audit log & retrace      │
  │ 3  │ Verification Line – Packet queue & inspect  │
  │ 4  │ Socket Identity – Socket registry           │
  │ 5  │ Flow Sorter      – Data volume analysis     │
  │ 6  │ Firewall Interfaces – Topology map          │
  │ 7  │ Validation Path – Shortest bypass route     │
  │ 8  │ Traffic Allocator – Load balancing          │
  │ 0  │ Exit                                        │
  └────┴─────────────────────────────────────────────┘
  Choice: 4

  [4] Socket Identity Registry
  a) Register socket   b) Query socket   c) Update status   d) List all
  > a
  Socket ID (e.g. SOCK-001): SOCK-001
  Source IP: 192.168.1.10
  Dest IP: 10.0.0.5
  Port (1-65535): 443
  Status (ACTIVE/BLOCKED/SUSPICIOUS): ACTIVE
  [SOCKET] Registered SOCK-001 | 192.168.1.10:443 → 10.0.0.5 | Status: ACTIVE
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard_Menu_Driven
  Choice: 5

  [5] Flow Sorter (Data Volume Analysis)
  a) Add flow   b) Show top N flows
  > a
  Flow ID (e.g. FLOW-001): FLOW-002
  Source IP: 192.168.1.12
  Dest IP: 8.8.8.9
  Data volume in bytes: 150000000
  [FLOW] Added flow FLOW-002 | 150000000 bytes

  ┌─────────────────────────────────────────────────┐
  │ MAIN MENU                                       │
  ├────┬────────────────────────────────────────────┤
  │ 1  │ Pattern List    — Manage attack signatures │
  │ 2  │ Logging Trail   — Audit log & retrace      │
  │ 3  │ Verification Line — Packet queue & inspect │
  │ 4  │ Socket Identity — Socket registry          │
  │ 5  │ Flow Sorter     — Data volume analysis     │
  │ 6  │ Firewall Interfaces — Topology map         │
  │ 7  │ Validation Path — Shortest bypass route    │
  │ 8  │ Traffic Allocator — Load balancing         │
  │ 0  │ Exit                                       │
  └────┴────────────────────────────────────────────┘
  Choice: 5

  [5] Flow Sorter (Data Volume Analysis)
  a) Add flow   b) Show top N flows
  > b
  Show top N flows (e.g. 5): 2

  ══ TOP 2 DATA FLOWS (by volume) ══
  Flow ID     Source          Destination     Volume
  --------------------------------------------------------
  FLOW-002    192.168.1.12    8.8.8.9         150000000 B ⚠ EXFIL RISK
  FLOW-001    192.168.1.10    8.8.8.8         50000000 B
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard_Menu_Driven
  Choice: 6

  [6] Firewall Interfaces (Topology)
  a) Add zone   b) Add link   c) Show topology
  > a
  Zone name (e.g. DMZ, INTERNAL, WAN, DB-ZONE): WAN
  [TOPO] Zone 'WAN' added.

  ┌──────────────────────────────────────────────────┐
  │  MAIN MENU                                       │
  ├────┬─────────────────────────────────────────────┤
  │ 1  │ Pattern List    — Manage attack signatures  │
  │ 2  │ Logging Trail   — Audit log & retrace       │
  │ 3  │ Verification Line — Packet queue & inspect  │
  │ 4  │ Socket Identity — Socket registry           │
  │ 5  │ Flow Sorter     — Data volume analysis      │
  │ 6  │ Firewall Interfaces — Topology map          │
  │ 7  │ Validation Path — Shortest bypass route     │
  │ 8  │ Traffic Allocator — Load balancing          │
  │ 0  │ Exit                                        │
  └────┴─────────────────────────────────────────────┘
  Choice: 6

  [6] Firewall Interfaces (Topology)
  a) Add zone   b) Add link   c) Show topology
  > b
  Zone A: DMZ
  Zone B: INTERNAL
  Link cost (default 1): 1
  [TOPO] Link: DMZ ↔ INTERNAL (cost=1)
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard
en
  MAIN MENU

1   Pattern List     — Manage attack signatures
2   Logging Trail    — Audit log & retrace
3   Verification Line — Packet queue & inspect
4   Socket Identity — Socket registry
5   Flow Sorter      — Data volume analysis
6   Firewall Interfaces — Topology map
7   Validation Path — Shortest bypass route
8   Traffic Allocator — Load balancing
0   Exit

Choice: 6

[6] Firewall Interfaces (Topology)
a) Add zone   b) Add link   c) Show topology
> c

══ FIREWALL TOPOLOGY MAP ══
[WAN] → (isolated)
[INTERNAL] → DMZ(c=1)
[DMZ] → INTERNAL(c=1)
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard
en

 MAIN MENU

 1   Pattern List      — Manage attack signatures
 2   Logging Trail     — Audit log & retrace
 3   Verification Line — Packet queue & inspect
 4   Socket Identity — Socket registry
 5   Flow Sorter       — Data volume analysis
 6   Firewall Interfaces — Topology map
 7   Validation Path — Shortest bypass route
 8   Traffic Allocator — Load balancing
 0   Exit

Choice: 7

[7] Validation Path (Shortest Bypass Route)
From zone: dmz
To zone  : INTERNAL
[PATH] Zone(s) not found.
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard

┌──────────────────────────────────────────────────────┐
│  MAIN MENU                                           │
├─────┬────────────────────────────────────────────────┤
│ 1   │ Pattern List    – Manage attack signatures     │
│ 2   │ Logging Trail   – Audit log & retrace          │
│ 3   │ Verification Line – Packet queue & inspect     │
│ 4   │ Socket Identity – Socket registry              │
│ 5   │ Flow Sorter     – Data volume analysis         │
│ 6   │ Firewall Interfaces – Topology map             │
│ 7   │ Validation Path – Shortest bypass route        │
│ 8   │ Traffic Allocator – Load balancing             │
│ 0   │ Exit                                           │
└─────┴────────────────────────────────────────────────┘
Choice: 8

[8] Traffic Allocator (Round-Robin Load Balancing)
a) Add server   b) Dispatch packet   c) Server status
> a
Server name (e.g. SEC-SERVER-01): SEC-SERVER-03
[ALLOC] Server 'SEC-SERVER-03' registered.
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard

┌──────────────────────────────────────────────────┐
│  MAIN MENU                                       │
├────┬─────────────────────────────────────────────┤
│ 1  │ Pattern List    — Manage attack signatures  │
│ 2  │ Logging Trail   — Audit log & retrace       │
│ 3  │ Verification Line — Packet queue & inspect  │
│ 4  │ Socket Identity — Socket registry           │
│ 5  │ Flow Sorter     — Data volume analysis      │
│ 6  │ Firewall Interfaces — Topology map          │
│ 7  │ Validation Path — Shortest bypass route     │
│ 8  │ Traffic Allocator — Load balancing          │
│ 0  │ Exit                                        │
└────┴─────────────────────────────────────────────┘
Choice: 8

[8] Traffic Allocator (Round-Robin Load Balancing)
a) Add server   b) Dispatch packet   c) Server status
> b
Packet ID to dispatch: PKT-001
[ALLOC] Packet PKT-001 → Server: SEC-SERVER-01
```

```
akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA % ./FireGuard

  MAIN MENU

  1 | Pattern List     — Manage attack signatures
  2 | Logging Trail    — Audit log & retrace
  3 | Verification Line — Packet queue & inspect
  4 | Socket Identity — Socket registry
  5 | Flow Sorter      — Data volume analysis
  6 | Firewall Interfaces — Topology map
  7 | Validation Path — Shortest bypass route
  8 | Traffic Allocator — Load balancing
  0 | Exit

Choice: 8

[8] Traffic Allocator (Round-Robin Load Balancing)
a) Add server   b) Dispatch packet   c) Server status
> c
Configured servers (3): SEC-SERVER-01 SEC-SERVER-02 SEC-SERVER-03
Next dispatch index: 0
```

```
┌──────────────────────────────────────────────────────┐
│  MAIN MENU                                           │
├────┬─────────────────────────────────────────────────┤
│ 1  │ Pattern List    — Manage attack signatures      │
│ 2  │ Logging Trail   — Audit log & retrace           │
│ 3  │ Verification Line — Packet queue & inspect      │
│ 4  │ Socket Identity — Socket registry               │
│ 5  │ Flow Sorter     — Data volume analysis          │
│ 6  │ Firewall Interfaces — Topology map              │
│ 7  │ Validation Path — Shortest bypass route         │
│ 8  │ Traffic Allocator — Load balancing              │
│ 0  │ Exit                                            │
└────┴─────────────────────────────────────────────────┘
  Choice: 0

  [FireGuard] Shutting down. All logs preserved. Stay secure.

akhilaanish2287@Akhilas-MacBook-Air DSA1-CASE-STUDY-AKHILA %
```